Kernel regression

- Definitions

Regressors in Bayesian regression

We have data $\left\{(x_n, y_n)\right\}_{n=1}^{N} \subset \mathbb{X} \times \mathbb{R}$ with scalar targets and seek a function

$$f : \mathbb{X} \mapsto \mathbb{R}$$

In Bayesian linear regression we use a model of the form

$$\begin{aligned} c &\sim \text{Normal}_M(\mu, C) \\ y_n | c &\sim \text{Normal}(\phi(x_n)c, \eta), \qquad n = 1, \ldots, N \end{aligned}$$

for a pre-chosen expansion $\phi : \mathbb{X} \mapsto \mathbb{R}^{1 \times M}$

Our regressor is defined indirectly

$$\begin{aligned} c &\sim \text{Normal}_M(\mu, C) \\ f(x) &= \phi(x)c, \qquad \forall x \in \mathbb{X} \end{aligned}$$

Regressors in Bayesian regression

Our regressor is defined indirectly

$$c \sim \text{Normal}_M(\mu, C)$$
$$f(x) = \phi(x)c, \qquad \forall x \in \mathbb{X}$$

We consider 1 point x_1

Our model provides the statistics

$$f = \left(f(x_1)\right) = \left[\phi(x_1)\right] c = \Phi c$$

$$f \sim \text{Normal}_1\left(\Phi\mu, \Phi C \Phi^t\right) = \text{Normal}_1(M, V)$$

We consider 2 points x_1, x_2

Our model provides the statistics

$$f = \begin{pmatrix} f(x_1) \\ f(x_2) \end{pmatrix} = \begin{bmatrix} \phi(x_1) \\ \phi(x_2) \end{bmatrix} c = \Phi c$$

$$f \sim \text{Normal}_2\left(\Phi\mu, \Phi C \Phi^t\right) = \text{Normal}_2(M, V)$$

We consider 3 points x_1, x_2, x_3

Our model provides the statistics

$$f = \begin{pmatrix} f(x_1) \\ f(x_2) \\ f(x_3) \end{pmatrix} = \begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \phi(x_3) \end{bmatrix} c = \Phi c$$

$$f \sim \text{Normal}_3\left(\Phi\mu, \Phi C \Phi^t\right) = \text{Normal}_3(M, V)$$

In this way, we can obtain the statistics of any discretization of the regressor

Regressors in Bayesian regression

In general, with N points $x_1, x_2, \ldots, x_N \in \mathbb{X}$, our model provides the statistics

$$f = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{pmatrix} = \begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \vdots \\ \phi(x_N) \end{bmatrix} c = \Phi c$$

$$\begin{aligned} f &\sim \text{Normal}_N\left(\Phi\mu, \Phi C \Phi^t\right) \\ &= \text{Normal}_N\left(M, V\right) \end{aligned}$$

This is the same as

$$\begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{pmatrix} \sim \text{Normal}_N\left(\begin{pmatrix} m(x_1) \\ m(x_2) \\ \vdots \\ m(x_N) \end{pmatrix}, \begin{bmatrix} v(x_1,x_1) & v(x_1,x_2) & \cdots & v(x_1,x_N) \\ v(x_2,x_1) & v(x_2,x_2) & \cdots & v(x_2,x_N) \\ \vdots & \vdots & \ddots & \vdots \\ v(x_N,x_1) & v(x_N,x_2) & \cdots & v(x_N,x_N) \end{bmatrix} \right)$$

with the mean and covariance defined by

$$m(x') = \phi(x')\mu \qquad\qquad v(x', x'') = \phi(x') C \phi^t(x'')$$

Regressors in Bayesian regression

In general, with N points $x_1, x_2, \ldots, x_N \in \mathbb{X}$, our model provides the statistics

$$\begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{pmatrix} \sim \text{Normal}_N \left(\begin{pmatrix} m(x_1) \\ m(x_2) \\ \vdots \\ m(x_N) \end{pmatrix}, \begin{bmatrix} v(x_1, x_1) & v(x_1, x_2) & \cdots & v(x_1, x_N) \\ v(x_2, x_1) & v(x_2, x_2) & \cdots & v(x_2, x_N) \\ \vdots & \vdots & \ddots & \vdots \\ v(x_N, x_1) & v(x_N, x_2) & \cdots & v(x_N, x_N) \end{bmatrix} \right)$$

with the mean and covariance given by

$$m(x') = \phi(x')\mu \qquad v(x', x'') = \phi(x')C\phi^t(x'')$$

Instead of starting with

$$\mu \in \mathbb{R}^M$$
$$C \in \mathbb{R}^{M \times M}_{sym,+}$$
$$\phi : \mathbb{X} \mapsto \mathbb{R}^{1 \times M}$$

to obtain $m(x')$ and $v(x', x'')$

a kernel regression starts directly with

$$m : \mathbb{X} \mapsto \mathbb{R}$$
$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

and, to ensure valid covariance matrices V, considers only symmetric positive semi-definite kernels v

Bayesian linear regression revisited

We have data $\left\{(x_n^\sharp, y_n^\sharp)\right\}_{n=1}^{N^\sharp} \subset \mathbb{X} \times \mathbb{R}$ with scalar targets and seek predictions at $\left\{x_n^\star\right\}_{n=1}^{N^\star} \subset \mathbb{X}$

We start with a mean and a symmetric positive semi-definite kernel

$$m : \mathbb{X} \mapsto \mathbb{R}$$
$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

and obtain the prior

$$\begin{pmatrix} f(x_1^\sharp) \\ \vdots \\ f(x_{N^\sharp}^\sharp) \\ f(x_1^\star) \\ \vdots \\ f(x_{N^\star}^\star) \end{pmatrix} \sim \text{Normal}_{N^\sharp + N^\star} \left(\begin{pmatrix} m(x_1^\sharp) \\ \vdots \\ m(x_{N^\sharp}^\sharp) \\ m(x_1^\star) \\ \vdots \\ m(x_{N^\star}^\star) \end{pmatrix}, \begin{bmatrix} v(x_1^\sharp, x_1^\sharp) & \cdots & v(x_1^\sharp, x_{N^\sharp}^\sharp) & v(x_1^\sharp, x_1^\star) & \cdots & v(x_1^\sharp, x_{N^\star}^\star) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v(x_{N^\sharp}^\sharp, x_1^\sharp) & \cdots & v(x_{N^\sharp}^\sharp, x_{N^\sharp}^\sharp) & v(x_{N^\sharp}^\sharp, x_1^\star) & \cdots & v(x_{N^\sharp}^\sharp, x_{N^\star}^\star) \\ v(x_1^\star, x_1^\sharp) & \cdots & v(x_1^\star, x_{N^\sharp}^\sharp) & v(x_1^\star, x_1^\star) & \cdots & v(x_1^\star, x_{N^\star}^\star) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v(x_{N^\star}^\star, x_1^\sharp) & \cdots & v(x_{N^\star}^\star, x_{N^\sharp}^\sharp) & v(x_{N^\star}^\star, x_1^\star) & \cdots & v(x_{N^\star}^\star, x_{N^\star}^\star) \end{bmatrix} \right)$$

Bayesian linear regression revisited

$$\begin{pmatrix} f(x_1^\sharp) \\ \vdots \\ f(x_{N^\sharp}^\sharp) \\ f(x_1^\star) \\ \vdots \\ f(x_{N^\star}^\star) \end{pmatrix} \sim \text{Normal}_{N^\sharp+N^\star} \left(\begin{pmatrix} m(x_1^\sharp) \\ \vdots \\ m(x_{N^\sharp}^\sharp) \\ m(x_1^\star) \\ \vdots \\ m(x_{N^\star}^\star) \end{pmatrix}, \begin{bmatrix} v(x_1^\sharp, x_1^\sharp) & \cdots & v(x_1^\sharp, x_{N^\sharp}^\sharp) & v(x_1^\sharp, x_1^\star) & \cdots & v(x_1^\sharp, x_{N^\star}^\star) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v(x_{N^\sharp}^\sharp, x_1^\sharp) & \cdots & v(x_{N^\sharp}^\sharp, x_{N^\sharp}^\sharp) & v(x_{N^\sharp}^\sharp, x_1^\star) & \cdots & v(x_{N^\sharp}^\sharp, x_{N^\star}^\star) \\ v(x_1^\star, x_1^\sharp) & \cdots & v(x_1^\star, x_{N^\sharp}^\sharp) & v(x_1^\star, x_1^\star) & \cdots & v(x_1^\star, x_{N^\star}^\star) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ v(x_{N^\star}^\star, x_1^\sharp) & \cdots & v(x_{N^\star}^\star, x_{N^\sharp}^\sharp) & v(x_{N^\star}^\star, x_1^\star) & \cdots & v(x_{N^\star}^\star, x_{N^\star}^\star) \end{bmatrix} \right)$$

Our likelihood take the form

$$\begin{pmatrix} y_1^\sharp \\ \vdots \\ y_{N^\sharp}^\sharp \end{pmatrix} \Bigg| \begin{pmatrix} f(x_1^\sharp) \\ \vdots \\ f(x_{N^\sharp}^\sharp) \end{pmatrix} \sim \text{Normal}_{N^\sharp} \left(\begin{pmatrix} f(x_1^\sharp) \\ \vdots \\ f(x_{N^\sharp}^\sharp) \end{pmatrix}, \eta \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \right)$$

Bayesian linear regression revisited

In more compact form, our mode is given by

$$\begin{pmatrix} f^\sharp \\ f^\star \end{pmatrix} \sim \text{Normal}_{N^\sharp + N^\star}\left(\begin{pmatrix} m^\sharp \\ m^\star \end{pmatrix}, \begin{bmatrix} v^{\sharp\sharp} & v^{\sharp\star} \\ v^{\star\sharp} & v^{\star\star} \end{bmatrix} \right)$$
$$y^\sharp \,\Big|\, f^\sharp \sim \text{Normal}_{N^\sharp}\left(f^\sharp, \eta I^{\sharp\sharp} \right)$$

Our model gives

$$f^\sharp \sim \text{Normal}_{N^\sharp}\left(m^\sharp, v^{\sharp\sharp} \right)$$
$$f^\star | f^\sharp \sim \text{Normal}_{N^\star}\left(m^\star + v^{\star\sharp}(v^{\sharp\sharp})^{-1}\left(f^\sharp - m^\sharp \right), v^{\star\star} - v^{\star\sharp}(v^{\sharp\sharp})^{-1} v^{\sharp\star} \right)$$
$$y^\sharp \,\Big|\, f^\sharp \sim \text{Normal}_{N^\sharp}\left(f^\sharp, \eta I^{\sharp\sharp} \right)$$

Bayesian linear regression revisited

Our model gives

$$
\begin{aligned}
f^{\sharp} &\sim \text{Normal}_{N^{\sharp}}\left(m^{\sharp}, v^{\sharp\sharp}\right) \\
f^{\star}|f^{\sharp} &\sim \text{Normal}_{N^{\star}}\left(m^{\star} + v^{\star\sharp}(v^{\sharp\sharp})^{-1}\left(f^{\sharp} - m^{\sharp}\right), v^{\star\star} - v^{\star\sharp}(v^{\sharp\sharp})^{-1}v^{\sharp\star}\right) \\
y^{\sharp}\,\Big|\,f^{\sharp} &\sim \text{Normal}_{N^{\sharp}}\left(f^{\sharp}, \eta I^{\sharp\sharp}\right)
\end{aligned}
$$

In turn, this gives

$$
\begin{aligned}
f^{\sharp}|y^{\sharp} &\sim \text{Normal}_{N^{\sharp}}\left(\bar{m}^{\sharp}, \bar{v}^{\sharp\sharp}\right) \\
f^{\star}|f^{\sharp} &\sim \text{Normal}_{N^{\star}}\left(m^{\star} + v^{\star\sharp}(v^{\sharp\sharp})^{-1}\left(f^{\sharp} - m^{\sharp}\right), v^{\star\star} - v^{\star\sharp}(v^{\sharp\sharp})^{-1}v^{\sharp\star}\right)
\end{aligned}
$$

with the updated hyper-parameters given by

$$
\bar{m}^{\sharp} = \bar{v}^{\sharp\sharp}\left((v^{\sharp\sharp})^{-1}m^{\sharp} + \frac{1}{\eta}y^{\sharp}\right), \qquad \bar{v}^{\sharp\sharp} = \left((v^{\sharp\sharp})^{-1} + \frac{1}{\eta}I^{\sharp\sharp}\right)^{-1}
$$

Bayesian linear regression revisited

Our model gives

$$f^\sharp | y^\sharp \sim \text{Normal}_{N^\sharp}\left(\bar{m}^\sharp, \bar{v}^{\sharp\sharp}\right)$$
$$f^\star | f^\sharp \sim \text{Normal}_{N^\star}\left(m^\star + v^{\star\sharp}(v^{\sharp\sharp})^{-1}\left(f^\sharp - m^\sharp\right), v^{\star\star} - v^{\star\sharp}(v^{\sharp\sharp})^{-1}v^{\sharp\star}\right)$$

Finally, this gives

$$f^\star | y^\sharp \sim \text{Normal}_{N^\star}\left(\bar{m}^\star, \bar{v}^{\star\star}\right)$$

with the hyper-parameters given by

$$\begin{aligned}
\bar{m}^\star &= m^\star + v^{\star\sharp}(v^{\sharp\sharp})^{-1}\left(\bar{m}^\sharp - m^\sharp\right), \\
\bar{v}^{\star\star} &= v^{\star\sharp}(v^{\sharp\sharp})^{-1}\bar{v}^{\sharp\sharp}(v^{\sharp\sharp})^{-1}v^{\sharp\star} + v^{\star\star} - v^{\star\sharp}(v^{\sharp\sharp})^{-1}v^{\sharp\star} \\
&= v^{\star\sharp}\left[(v^{\sharp\sharp})^{-1}\bar{v}^{\sharp\sharp}(v^{\sharp\sharp})^{-1} - (v^{\sharp\sharp})^{-1}\right]v^{\sharp\star} + v^{\star\star}
\end{aligned}$$

Bayesian linear regression revisited

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$$f^\sharp | y^\sharp \sim \text{Normal}_{N^\sharp}\left(\bar{m}^\sharp, \bar{v}^{\sharp\sharp}\right)$$
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Finally, this gives

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&= v^{\star\sharp}\left[(v^{\sharp\sharp})^{-1}\bar{v}^{\sharp\sharp}(v^{\sharp\sharp})^{-1} - (v^{\sharp\sharp})^{-1}\right]v^{\sharp\star} + v^{\star\star} \\
&= v^{\star\sharp}(v^{\sharp\sharp})^{-1}\left[\bar{v}^{\sharp\sharp}(v^{\sharp\sharp})^{-1} - I^{\sharp\sharp}\right]v^{\sharp\star} + v^{\star\star}
\end{aligned}$$

Kernels

To conduct kernel regression, we need a mean and a kernel

$$m : \mathbb{X} \mapsto \mathbb{R}$$
$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

The kernel must be symmetric

$$v(x', x'') = v(x'', x'), \qquad \forall x', x'' \in \mathbb{X}$$

and positive semi-definite, this means that

$$\sum_{n'=1}^{N} \sum_{n''=1}^{N} \gamma_{n'} \gamma_{n''} v(x_{n'}, x_{n''}) \geq 0$$

for any choice of $x_1, \ldots, x_N \in \mathbb{X}, \gamma_1, \ldots, \gamma_N \in \mathbb{R}$ and $N = 1, 2, \ldots$

- Having a symmetric positive semi-definite kernel v_0, new kernels can be obtained by

$$v(x', x'') = c v_0(x', x''), \qquad c > 0$$
$$v(x', x'') = c + v_0(x', x''), \qquad c > 0$$

- Having symmetric positive semi-definite kernels v_1 and v_2, new kernels can be obtained by

$$v(x', x'') = v_1(x', x'') + v_2(x', x'')$$
$$v(x', x'') = v_1(x', x'') v_2(x', x'')$$